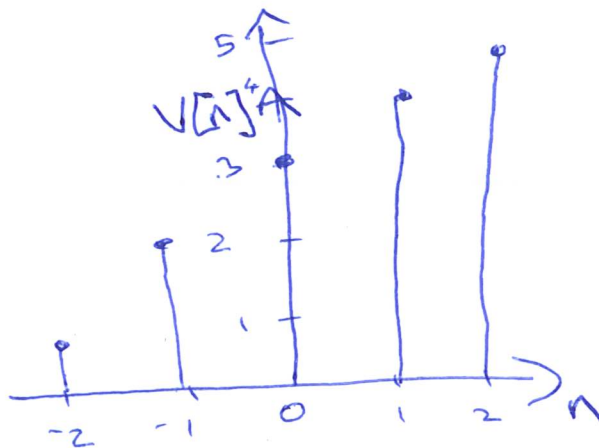


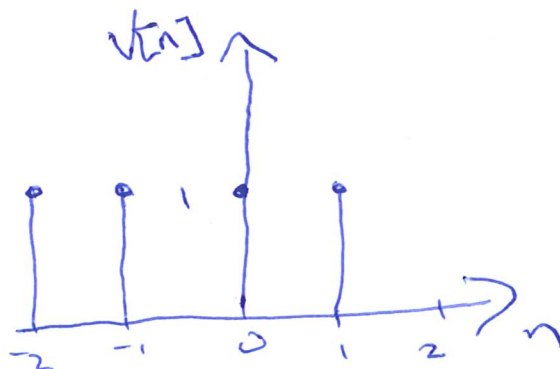
ENMT301 - Signals & Sensors Tutorial 1 Solutions

- Q1)
- a) 3
 - b) 6
 - c) 7
 - d) 8
 - e) 11
 - f) 10
 - g) 9
 - h) 1
 - i) 14
 - j) 15
 - k) 16
 - l) 12
 - m) 13
 - n) 17
 - o) 4
 - p) 2
 - q) 5
 - r) 18

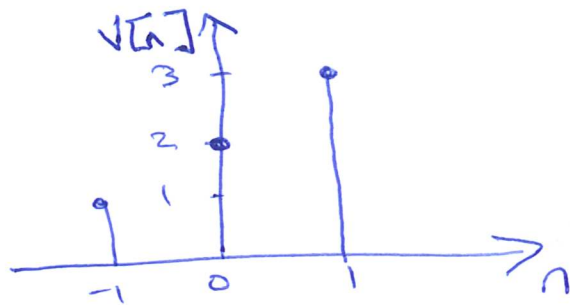
2) a)



b)



2) c)



$$d) v[n] = \delta[n+2] + 2\delta[n+1] + 3\delta[n] + 4\delta[n-1] + 5\delta[n-2]$$

$$e) v[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2] + 4\delta[n-3] + 5\delta[n-4]$$

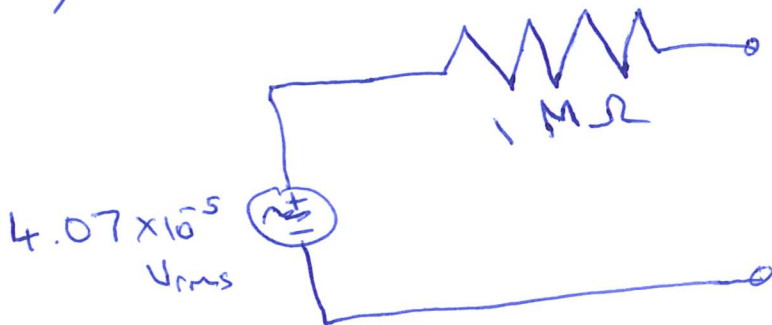
3) a)

$$V_n = \sqrt{4kTRB}$$

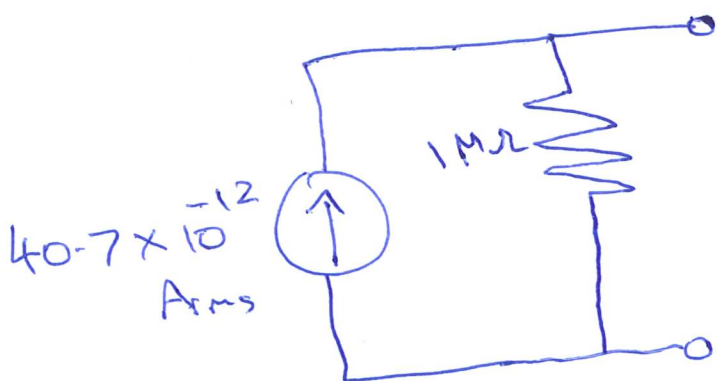
$$V_n = \sqrt{4 \times 1.38 \times 10^{-23} \times 300 \times 1 \times 10^6 \times 100 \times 10^3}$$

$$V_n = 4.07 \times 10^{-5} \text{ V}_{rms}$$

b) Thévenin equivalent circuit



Norton equivalent circuit



$$I_{no} = \frac{V_{th}}{R}$$

$$I_{no} = \frac{4.07 \times 10^{-5}}{1 \times 10^6}$$

$$I_{no} = 40.7 \text{ pA}_{rms}$$

3) c)



$$\frac{1}{R_{tot}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{tot} = \frac{1 \times 10^6 \times 1 \times 10^3}{1 \times 10^6 + 1 \times 10^3}$$

$$R_{tot} = 999 \Omega$$

$$V_n = \sqrt{4kT R_{tot} B} = \sqrt{4 \times 1.38 \times 10^{-23} \times 300 \times 999 \times 100 \times 10^3}$$

$$V_n = 1.29 \times 10^{-6} V_{rms}$$



Noise is uncorrelated so can add noise sources or resistances

$$R_{tot} = R_1 + R_2 = 1 \times 10^3 + 1 \times 10^6 = 1.001 \times 10^6 \Omega$$

$$V_n = \sqrt{4kT R_{tot} B}$$

$$V_n = \sqrt{4 \times 1.38 \times 10^{-23} \times 300 \times 1.001 \times 10^6 \times 100 \times 10^3}$$

$$V_n = 4.07 \times 10^{-5} V_{rms}$$